

Regular Expression to DFA Conversion and C++ Implementation

Regular Expression (RE):

RE = **b* a [a + b a (a + b)]***

Project Description

This project demonstrates the complete process of converting a **Regular Expression (RE)** into a **Deterministic Finite Automaton (DFA)** and implementing it in **C++**.

The following steps are covered:

1. Thompson's Construction

- The given regular expression is first converted into an **ϵ -NFA (Epsilon Non-Deterministic Finite Automaton)** using Thompson's Construction method.

2. ϵ -Closure Computation

- The **ϵ -closure** of each state in the ϵ -NFA is calculated to handle epsilon transitions properly.

3. NFA to DFA Conversion

- The ϵ -NFA is transformed into an equivalent **DFA** using the **subset construction algorithm**.

4. DFA Implementation in C++

- The final DFA is implemented in **C++** to simulate string acceptance based on the given regular expression.

Objective

The main goal of this project is to understand:

- Regular Expressions
- Thompson's Construction
- ϵ -NFA and ϵ -Closure
- NFA to DFA conversion
- Practical DFA implementation using C++

